

An Analytical Review of Provisions for Visually Impaired Persons in Indian Higher Education: A Point-by-Point Extraction from the UGC Accessibility Guidelines and Standards

Executive Summary & Comprehensive Matrix of Provisions

This report provides a definitive, point-by-point analysis of all provisions for visually impaired individuals, including those with blindness and low vision, as mandated by the University Grants Commission's (UGC) "Accessibility Guidelines and Standards for Higher Education Institutions and Universities".¹ The guidelines, issued in June 2022 and subsequently incorporated into the Rights of Persons with Disabilities Rules, 2017, represent a foundational shift in regulatory expectations for Higher Education Institutions (HEIs) in India. The document moves beyond rudimentary accommodations to mandate a holistic and integrated ecosystem of accessibility, encompassing institutional policy, physical infrastructure, digital resources, academic frameworks, and campus life.

The analysis is structured to be strictly faithful to the source document, extracting every relevant guideline and standard. The objective is to create a clear, structured, and granular knowledge base suitable for policy implementation, compliance auditing, and integration into advanced information retrieval systems. The comprehensive matrix below serves as a quick-reference tool, consolidating all specific provisions for visually impaired persons and mapping them to the relevant guideline category and document reference. This structured format transforms scattered data points into a relational database, allowing for efficient and targeted information access.

Comprehensive Matrix of Provisions for Visually Impaired Individuals

Guideline Category	Specific Provision	Target Group	Document Reference
Foundational Needs	Access to information via Braille, audio, enlarged print, digital formats with screen readers, and tactile systems.	Both	Sec. 2.1.3, p. 16
Foundational Needs	Assistance with orientation and mobility (O&M).	Both	Sec. 2.1.3, p. 16

Foundational Needs	Support in concept development.	Both	Sec. 2.1.3, p. 16
Foundational Needs	Provision of emotional-behavioral and psycho-social support.	Both	Sec. 2.1.3, p. 16
Administrative Support	Provision of a "Plus-Curriculum" including Braille reading/writing, O&M, and independent living skills.	Both	Sec. 2.4, p. 26
Administrative Support	Training in sensory efficiency skills (use of hearing, touch) and use of optical devices.	Low Vision	Sec. 2.4, p. 26
Administrative Support	Instruction in the use of mathematical devices like the Taylor's Frame and Abacus.	Blindness	Sec. 2.4, p. 26
Teaching & Learning	Allowing students to determine ideal seating location to see, hear, or touch material.	Both	Sec. 2.5, p. 29
Teaching & Learning	Verbal description of visual class activities (e.g., show of hands).	Both	Sec. 2.5, p. 29
Teaching & Learning	Permission to use voice recorders, computers, or digital slates for note-taking.	Both	Sec. 2.5, p. 29
Examinations	Provision of compensatory (extra) time.	Both	Sec. 2.6, p. 31
Examinations	Question papers in alternate formats: Braille, large print, audio, or with a reader.	Both	Sec. 2.6, p. 31; Sec. 7.3.2, p. 84
Examinations	Permission to use computers or other assistive technology during exams.	Both	Sec. 2.6, p. 31; Sec. 7.3.4, p. 84

Examinations	Use of specific assistive devices like tailor frame, Braille slate, abacus, and geometry kit.	Both	Sec. 7.3.4, p. 84
Examinations	Provision of alternate objective questions in lieu of questions requiring visual inputs (e.g., graphs).	Both	Sec. 7.3.7, p. 84
ICT & Digital Resources	Procurement of ICT must adhere to BIS IS 17802; biometric systems must have speech/haptic output.	Both	Sec. 3.1, p. 38
ICT & Digital Resources	Digital documents (PDFs) must conform to PDF-UA (Universal Accessibility) standards.	Both	Sec. 3.2.1, p. 40
ICT & Digital Resources	Websites and videos must conform to WCAG Level AA standards.	Both	Sec. 3.2.1, p. 40
ICT & Digital Resources	Audio materials should be in DAISY (Digital Accessible Information System) format.	Both	Sec. 3.2.1, p. 40
ICT & Digital Resources	HEIs must provide assistive technology like computers with screen readers and refreshable Braille displays.	Both	Sec. 3.4, p. 43
Learning Resources	HEIs must use and contribute to the Sugamya Pustakalaya national online library of accessible books.	Both	Sec. 3.2.4, p. 42
Learning Resources	Libraries must provide services including talking books, Braille materials, and large-print books.	Both	Sec. 8.3.2, p. 93
Mobility & Wayfinding	Installation of continuous tactile pathways connecting all key buildings.	Blindness	Sec. 4.3.1, p. 49; Sec. 5.1.3, p. 57

Mobility & Wayfinding	Use of GPS mapping or Bluetooth beacon-based technologies for campus navigation.	Both	Sec. 4.4, p. 50
Mobility & Wayfinding	Provision of tactile maps or models for orientation.	Blindness	Sec. 5.11, p. 66
Built Infrastructure	Walkway lighting at 35-40 lux to ensure visibility of tactile pavers for persons with low vision.	Low Vision	Sec. 5.1.8, p. 58
Built Infrastructure	Classroom lighting at 250-300 lux (general) and 500-750 lux (teacher's area), avoiding flicker and glare.	Low Vision	Sec. 5.2.4, p. 59
Built Infrastructure	Stair edges must have a 40mm color-contrasting band.	Low Vision	Sec. 5.2.6, p. 61
Built Infrastructure	Tactile warning pavers must be installed 300mm before and after stairs and ramps.	Blindness	Sec. 5.2.6, p. 61
Built Infrastructure	Lift controls must have Braille and tactile markings, with audio-visual announcements.	Both	Sec. 5.2.6, p. 61
Built Infrastructure	Auditorium row and seat numbers must be in Braille and tactile formats.	Both	Sec. 5.3, p. 62
Campus Amenities	ATMs must be Braille-enabled with a headphone jack for audio output.	Both	Sec. 5.10, p. 65
Emergency Systems	Emergency alarms must be both audio (hooter) and visual (flashing bulb).	Both	Sec. 5.12, p. 66

Section I: Foundational Needs and Support Provisions

The guidelines establish a comprehensive framework for support that begins with a detailed assessment of the specific needs of visually impaired students. This foundational

understanding serves as the justification for all subsequent technical and procedural mandates within the document. The provisions detailed in Chapter 2 create a clear causal link between a student's requirements and the institution's responsibilities, ensuring that every mandated facility or accommodation is a direct response to a recognized need.¹

1.1 Disability-Specific Needs: Blindness and Low Vision

Section 2.1.3 of the guidelines provides a granular breakdown of the needs of students with blindness and low vision, which HEIs are required to address.¹ These needs extend beyond academics to encompass mobility, psycho-social well-being, and daily functioning on campus.

- **Information Access Modalities:** The primary need identified is the ability to access information through non-visual or enhanced-visual means. The document explicitly lists a variety of required formats: Braille, audio, enlarged print, and digital formats that are compatible with screen reading software. It also includes a broader category of "other tactile and sensory systems," acknowledging the diverse technologies and methods used.¹
- **Orientation and Mobility (O&M):** A critical need for assistance with orientation (understanding one's position in an environment) and mobility (navigating that environment safely and independently) is specified. This acknowledges that navigating a large and unfamiliar university campus presents significant challenges.¹ The technical standards for tactile pathways and GPS mapping in later chapters are a direct response to this identified need.
- **Conceptual Development:** The guidelines recognize that students with visual impairments, particularly those with congenital blindness, may require support in concept development. This is a nuanced acknowledgement that incidental learning, which often occurs visually, may be limited, and formal instruction may be needed to build a strong foundation for abstract academic concepts.¹
- **Psycho-Social and Emotional Support:** The document mandates that HEIs recognize and provide for the potential need for emotional-behavioral and psycho-social support. This addresses the holistic well-being of the student, acknowledging that adapting to a higher education environment can present unique social and emotional challenges.¹
- **Infrastructure and ICT Assistance:** A general need for assistance in the use of both physical campus infrastructure and Information and Communication Technology (ICT) is listed. This establishes the basis for providing hands-on training and support for navigating buildings and using university portals, software, and other digital tools.¹
- **Medical Needs:** The guidelines also include the possibility of frequent therapy or medication as a specific need, ensuring that the institution's support systems are prepared to accommodate health-related requirements.¹

1.2 Administrative and Academic Support: The Plus-Curriculum Framework

The guidelines introduce a transformative concept by mandating a "Plus-Curriculum" or "Expanded Core-Curriculum" for students with visual disabilities.¹ This represents a significant evolution in the role of an HEI, shifting its responsibility from merely providing academic content to also ensuring students possess the foundational skills necessary to access that content. This framework mandates that HEIs provide structured training in areas that are prerequisites for academic success and independent living.

This approach demonstrates a sophisticated understanding that true access is not achieved by simply placing a student in a classroom. It requires equipping the student with the specific tools and skills to engage with the curriculum on an equal basis. By mandating this plus-curriculum, the UGC effectively redefines the scope of an HEI's duty of care, compelling them to establish resource centers and hire specialized personnel, such as O&M instructors and Braille experts, to deliver this essential training.

The specific components of the Plus-Curriculum include ¹:

- **Braille Literacy:** Formal instruction in both Braille Reading and Braille Writing.
- **Orientation & Mobility (O&M):** Professional training in O&M techniques for navigating the campus and community.
- **Sensory and Daily Living Skills:** This includes training in Independent Living Skills and Sensory Training, which focuses on the efficient use of hearing, touch, and other senses. For students with low vision, this also includes training in the use of optical devices.
- **Specialized Academic Tools:** Instruction in the use of non-visual mathematical devices, specifically the **Taylor's Frame & Abacus**.
- **Compensatory Skills:** Development of a range of skills necessary for accessing the core curriculum, such as concept development, organizational and study skills, the interpretation of tactile graphics, and the effective use of audio materials.
- **Assistive Technology (AT):** Comprehensive training in the use of relevant assistive technologies.

1.3 Teaching and Learning Accommodations

Section 2.5 of the guidelines details specific accommodations required within the teaching and learning environment to ensure the full participation of students with visual disabilities.¹ These measures focus on adapting the physical classroom and pedagogical methods.

- **Classroom Environment:** Students must be allowed to determine their own ideal seating location. This empowers the student to choose a position that best allows them to see, hear, or, if necessary, touch the presented material, accommodating the wide variance in individual needs.
- **Verbalization of Visual Information:** A crucial pedagogical adaptation is the requirement for faculty to provide verbal descriptions of visual information and activities. The example given—stating how many hands were raised in response to a

question—illustrates the principle of ensuring that non-visual learners are not excluded from the interactive elements of a lecture.

- **Laboratory Support:** In scientific and technical courses, HEIs must provide assistance in labeling lab materials so that they are easily identifiable through tactile or large-print means.
- **Note-Taking Support:** The guidelines explicitly permit the use of various technologies for note-taking, including voice recorders, computers, and digital slates with styluses, recognizing that traditional pen-and-paper methods may not be accessible.

1.4 Assessment and Examination Accommodations

To ensure that examinations test a student's knowledge of a subject rather than their ability to overcome the limitations of their disability, the guidelines mandate a series of accommodations for assessments and examinations.¹

- **Compensatory Time:** The provision of compensatory, or extra, time for completing examinations is a standard requirement.
- **Alternate Formats:** Question papers must be made available in a range of accessible formats to suit the student's needs. These include **Braille, large print, and audio**. The use of a human reader is also explicitly permitted.
- **Technology in Exams:** Students must be allowed to use computers or other assistive technology during examinations to read questions and write answers.

Section II: Accessible Information, Communication, and Learning Resources

The guidelines dedicate significant attention to ensuring that all information within the HEI ecosystem is accessible, from procurement policies for technology to the format of daily instructional materials. This section signals a systemic shift, pushing institutions away from a reactive model of converting materials on demand towards a proactive model where accessibility is a core requirement from the point of creation or purchase.

2.1 Procurement of Accessible ICT

Section 3.1 establishes a clear and non-negotiable standard for the procurement of all Information and Communication Technology (ICT).¹ This upstream focus is a key policy lever, designed to solve accessibility issues at their source rather than through costly downstream remediation. By embedding accessibility into the procurement process, the guidelines aim to make it a default feature of the campus's technological infrastructure.

- **Mandatory Adherence to Standards:** All ICT products and services procured or created by HEIs must adhere to the standards notified by the Bureau of Indian Standards (BIS), specifically IS 17802. This includes everything from university websites and learning management systems to software and office equipment.
- **Principle of Non-Exclusion:** The policy is anchored by an unambiguous directive: an institution **"shall not purchase any ICT devices or services that are not usable by persons with disabilities"**.¹ This clause has profound implications for institutional purchasing departments, requiring them to develop expertise in accessibility standards and to include accessibility compliance as a mandatory criterion in all tenders. For example, a biometric attendance system would be non-compliant unless it provides **speech or haptics output** for users with blindness.¹

2.2 Accessibility of Instructional Materials

The guidelines stress the importance of creating content that is "Born Accessible," meaning it is designed for accessibility from the outset.¹ This approach is more efficient and provides a better user experience than retrofitting inaccessible documents.

- **Digital Format Standards:** To ensure interoperability with assistive technologies, the guidelines specify the technical standards that various digital formats must meet ¹:
 - **PDF:** Documents must conform to the PDF-UA (Universal Accessibility) standard, which ensures a logical reading order and proper tagging for screen readers. A simple scanned image of a document is explicitly identified as inaccessible.
 - **Websites and Videos:** All web content, including videos, must conform to the Web Content Accessibility Guidelines (WCAG) at Level AA.
 - **Audio:** Audio-based instructional materials should be provided in the DAISY (Digital Accessible Information System) format, which allows for structured navigation similar to a print book.
 - **Fonts:** For Indian languages, digital documents must use **Unicode-compliant fonts** to ensure they can be correctly interpreted and read aloud by screen reading software.¹
- **Conversion Services:** While promoting the "Born Accessible" principle, the guidelines acknowledge the vast amount of existing inaccessible material. HEIs are therefore held responsible for converting instructional materials like print books and photocopied handouts into accessible formats, including **accessible EPUB, Braille, large print, and audio**. This responsibility is legally supported by Section 52(1)(zb) of the Indian Copyright Amendment Act (2012), which permits such conversion for persons with disabilities without requiring prior permission from the copyright holder.¹

2.3 Sugamya Pustakalaya: The National Online Library

The mandate to integrate with the **Sugamya Pustakalaya** (www.sugamypustakalaya.in) is a cornerstone of the guidelines' strategy for providing accessible texts.¹ This requirement moves beyond institutional self-sufficiency to create a collaborative, national ecosystem for accessible learning resources. By mandating participation, the UGC prevents the creation of hundreds of redundant, isolated collections of accessible books, instead fostering a federated, crowd-sourced national asset. This approach represents a highly efficient use of public and institutional resources, as a book converted by one university becomes immediately available to all others, creating a powerful network effect that rapidly expands the availability of materials.

- **Centralized Repository:** HEIs are required to use the Sugamya Pustakalaya, a shared online library of accessible format books for persons with print disabilities.
- **Mandatory Workflow:** The guidelines prescribe a specific workflow: before undertaking the resource-intensive task of converting a book, an institution must first search the repository. If the book is not available and is subsequently converted, the institution is then obligated to upload the new accessible version to the Pustakalaya, thereby enriching the national collection.
- **Membership Provision:** HEIs must provide membership to the Sugamya Pustakalaya to all their students and faculty members who have print disabilities.

2.4 Provision of Assistive Technologies (AT)

The guidelines recognize that access to information is contingent on access to the right technology. As part of the principle of reasonable accommodation, HEIs have a direct responsibility to provide students with the necessary Assistive Technology (AT) devices.¹

- **Institutional Responsibility:** HEIs must ensure that every student, faculty, or employee with a disability is provided with the AT they require. The document provides specific examples, such as a **computer equipped with screen reading software, a refreshable Braille display, and DAISY players**.¹
- **Training and Awareness:** The provision of a device is considered incomplete without proper support. The guidelines mandate that the device must be accompanied by **appropriate training** for the user to ensure they can fully utilize its capabilities. Furthermore, HEIs must organize awareness and orientation events for faculty and staff about these technologies, so that their use can be effectively integrated into teaching and learning methodologies.¹

Section III: The Accessible Campus: Mobility, Infrastructure, and Wayfinding

This section synthesizes the detailed technical specifications for the physical campus environment, ensuring it is navigable, safe, and usable for students with visual impairments. The guidelines mandate a multi-modal and redundant design philosophy, combining physical, tactile, and digital systems to create a robust wayfinding ecosystem. A notable feature of this

section is its nuanced approach, providing differentiated standards that cater to the distinct needs of individuals who are blind and those who have low vision.

3.1 Inclusive Mobility and Orientation

Chapter 4 of the guidelines focuses on creating an environment that supports independent mobility and orientation for all students.¹

- **Tactile Pathways:** The installation of a **tactile guidance system** is a mandatory feature for pedestrian pathways. These paths, which use textured ground surface indicators, must be continuous, free of breaks, and form a logical network connecting all major buildings, facilities, and transport points on campus. This provides a physical, non-visual route for individuals who are blind to follow.¹
- **Technological Aids for Navigation:** Recognizing the power of modern technology, the guidelines strongly encourage HEIs to supplement physical infrastructure with digital navigation aids. The use of **GPS mapping of the campus or the deployment of Bluetooth beacon-based technologies** is recommended. These systems allow students with visual impairments to use their smartphones with screen readers to get real-time, turn-by-turn directions to locate buildings, classrooms, and other services.¹
- **Pathway and Surface Standards:** To ensure safety for all, including those with low vision and mobility impairments, walkways must be at least 1800 mm wide to allow for two-way traffic. The surface must be firm, level, and non-slip. A minimum vertical clearance of 2100 mm must be maintained along all accessible routes, free of any protruding obstacles like low-hanging branches or signs.¹

3.2 Signage and Wayfinding Systems

The guidelines mandate a layered, multi-modal approach to signage, ensuring that information is available through multiple sensory channels.¹ This redundancy ensures that if one method of accessing information fails or is unsuitable for a user, others are available.

- **Multi-Modal Information:** All key signage must provide information in visual, audible, and **tactile forms**. Tactile information includes signs with **embossed (raised) lettering or Braille**.¹
- **Tactile Maps:** To provide a spatial overview of the campus or building layouts, HEIs should install **tactile maps or models**. These allow users with visual impairments to understand the relative location of different facilities and plan their routes in advance.¹
- **Technical Specifications for Visibility:** For the benefit of individuals with low vision, signage must be designed with high visual contrast. The material should be non-reflective with a matt finish to prevent glare, and it must be adequately illuminated to ensure legibility in all lighting conditions.¹

3.3 Lighting Standards for Low Vision

The guidelines demonstrate a sophisticated understanding of visual impairment by including specific standards for illumination, which are critical for students with low vision.

- **Pathway Illumination:** Outdoor walkways, especially those with tactile pavers, should be illuminated with white lighting at an average level of **35-40 lux**. This is specified to ensure that the color contrast of the tactile indicators remains visible at night to persons with low vision, aiding in navigation.¹
- **Classroom Illumination:** Inside learning spaces, a general illumination level of **250-300 lux** is recommended. To help students with low vision focus on the instructor, a higher illumination level of **500-750 lux** is preferable in the area where the teacher is located. The lighting systems must be designed to avoid flicker and glare, which can cause eye strain and headaches.¹

3.4 Specific Infrastructure Elements

The guidelines extend accessibility requirements to numerous specific elements of the built environment throughout the campus.

- **Stairs:** To make level changes safer and more discernible for people with low vision, the edge of each stair must be marked with a **40mm color-contrasting band**. For individuals who are blind, **tactile warning pavers** must be installed 300mm before the start of a stair run and 300mm after the end.¹ A staircase is only considered fully compliant when it addresses both sets of needs.
- **Lifts/Elevators:** All lift control panels, both inside the car and in the lobby, must have **Braille and tactile markings**. To provide non-visual information about the lift's movement, there must be **audio-visual announcements** indicating the floor number and direction of travel.¹
- **Assembly Areas/Auditoriums:** To aid in finding seats in large venues, the row and seat number identification signs must be provided in **Braille and tactile formats**.¹
- **Campus Amenities (ATMs):** Any Automated Teller Machines (ATMs) installed on campus must be accessible. This includes being **Braille-enabled** and having a standard headphone jack to provide **audio output** of on-screen information for private use.¹
- **Emergency Preparedness:** Emergency alarm systems must be accessible to all. This requires them to be both **audio (a loud hooter or siren)** and **visual (a bright, flashing strobe light)**, ensuring that the alarm is perceived by individuals with vision and hearing disabilities.¹

Section IV: Academic and Examination Frameworks

The core academic functions of an HEI—curriculum delivery and student assessment—are subject to rigorous accessibility standards under the guidelines. The overarching principle is to ensure that all students have an equal opportunity to learn and to demonstrate their learning. The examination provisions, in particular, are designed to ensure that assessments are a valid measure of a student's subject knowledge, not a test of their ability to function with a disability.

4.1 Accessible Curriculum Design and Delivery

Chapter 6 of the guidelines outlines the requirements for making the curriculum and its associated learning resources accessible.¹ The document promotes a dual approach: a proactive one, where courses are designed accessibly from the start using principles of Universal Design for Learning (UDL), and a reactive one, where individual accommodations are provided as needed. The emphasis is on shifting institutional practice towards the proactive model.

- **Alternate Formats:** An accessible curriculum requires that all instructional materials are available in formats that students with visual impairments can use. The guidelines explicitly mention the conversion of print materials into **Braille, audio, or large print**, and the provision of **captioning and audio descriptions** for video materials.¹
- **Screen Reader Compatibility:** The guidelines highlight a common technical barrier, stating that digital documents, such as PDFs, must not be mere scanned images. Such files are inaccessible because **screen reader technology cannot interpret the text**. All digital content must be properly formatted, text-based, and accessible.¹
- **Accessible Library Resources:** The campus library is identified as a critical hub for accessible resources. Libraries are required to actively procure and provide materials in accessible formats. This includes maintaining a **"print-accessible library"** with the capacity to make any content accessible upon request. Specific resources mentioned include **talking books, Braille materials, and large-print books**.¹

4.2 Accessible Assessment and Examination Systems

Chapter 7 reiterates and expands upon existing UGC provisions to create a robust framework for accessible examinations.¹ These measures are designed to remove barriers in the assessment process, ensuring that students are evaluated on their academic merit alone. This requires examination departments to move beyond logistics and engage in the pedagogical redesign of assessments to create equivalent, non-visual evaluation methods.

- **UGC Provisions:** The guidelines incorporate the provisions of UGC Notification F.No. 6-2/2013 (SCT), which mandate the following¹:

- **Mode of Examination:** Students with benchmark disabilities must be given the option to take examinations in their preferred format, including **Braille, on a computer, or in large print**.
- **Assistive Devices:** Candidates must be permitted to use their standard assistive devices in the examination hall. For students with visual impairments, this includes tools like the **tailor frame, Braille slate, abacus, geometry kit, and Braille measuring tape**.
- **Alternate Question Design:** This is a critical pedagogical requirement. For questions that rely on visual inputs, such as interpreting a graph, chart, or image, the examining body must provide **alternate questions** that assess the same underlying skill or knowledge without relying on vision.
- **Open Book Exams:** In the context of an open book examination, the required reading materials must be provided to the student in an accessible format, such as **Braille or an e-text document on a computer equipped with suitable screen reading software**.
- **Flexibility in Digital Assessment:** The guidelines also look towards modern assessment methods, promoting flexibility in digital exams. This includes allowing students to personalize the view of questions (e.g., changing font type, size, or colors) and providing built-in **text-to-speech software to read questions aloud**, a feature that benefits students with vision disabilities as well as those with learning disabilities like dyslexia.¹